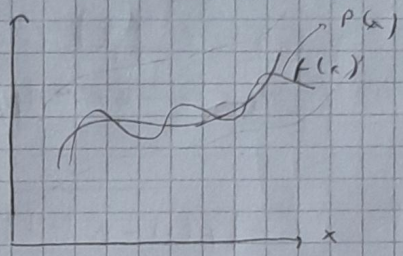
İnterpolasyon ve Polinomlar ile yaklaşım

Teorem: (Weierstrass yaklaşım Teoremi)

$f \in C[a,b]$ → keyfi bir $\varepsilon > 0$ için öyle bir $P(x)$ bulunur ki, her $x \in [a,b]$ için $|f(x) - P(x)| < \varepsilon$

$$f(x) - \varepsilon < P(x) < f(x) + \varepsilon$$



Lagrange IP

$y_0 = f(x_0)$

$y_1 = f(x_1)$

$$P(x) = \underbrace{\frac{x-x_1}{x_0-x_1}}_{L_0(x)} \overset{f(x_0)}{y_0} + \underbrace{\frac{x-x_0}{x_1-x_0}}_{L_1(x)} \overset{f(x_1)}{y_1}$$

$$P_n(x) = L_{n,0}(x) f(x_0) + L_{n,1}(x) f(x_1) + \cdots + L_{n,n}(x) f(x_n)$$

$$L_{n,k}(x) = \frac{(x-x_0)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)}{(x_k-x_0)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n)} = \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{(x-x_i)}{(x_k-x_i)}$$

$L_{n,k}(x_i) = 0 \quad i \neq k$

$L_{n,k}(x_k) = 1$

$n=2 \quad P_2(x) = \underbrace{L_{2,0}(x)}_{\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}} f(x_0) + \underbrace{L_{2,1}(x)}_{\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x-x_2)}} f(x_1) + \underbrace{L_{2,2}(x)}_{\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}} f(x_2)$

Örnek 1 $f(x) = \frac{1}{x}$

a) $x_0 = 2$, $x_1 = \frac{11}{4}$, $x_2 = 4$ olduğunu noktaları için 2. mertebe LIP

b) $f(3) = \frac{1}{3}$ için LIP ile bulunuz

x	$f(x)$
$x_0 = 2$	$1/2$
$x_1 = 11/4$	$4/11$
$x_2 = 4$	$1/4$

$$L_{2,0}(x) = \frac{2}{3}\left(x - \frac{1}{4}\right)(x-4)$$

$$L_{2,1}(x) = -\frac{16}{5}(x-2)(x-4)$$

$$L_{2,2}(x) = \frac{2}{5}(x-2)\left(x - \frac{11}{4}\right)$$

$$P_2(x) = \frac{1}{22}x^2 - \frac{35}{88}x + \frac{49}{44}$$

$P_2(3) \cong 0,3295$

Teorem $x_0, x_1, \dots, x_n$ $[a,b]$'de $n+1$ adet farklı noktadır

$f \in C^{n+1}[a,b]$ olsun

$$f(x) = P_n(x) + \underbrace{\frac{f^{n+1}(\xi)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)}_{\text{Hata}}$$

Örnek

$x \in [2,4]$ $f(x) = \frac{1}{x}$ fonk

örnek1. aşağıdaki noktalarının 2. mertebeden LP'i nolası

$$R(x) = \frac{f'''(\xi)}{3!}(x-2)(x-\frac{11}{4})(x-4) \quad \xi \in [2,4]$$

$\left(f'''(x) = -6x^{-4} \;/\!/\; -\frac{g(x)}{\xi^4}\right.$

$$|f(x) - p_2(x)| = |R(x)| \le \frac{1}{16} \cdot \frac{9}{10} \approx 0{,}0352$$

$g(x) \Rightarrow$ kritik x_1

$g'(x) = 0$

$x = \frac{3}{2}$ $x = \frac{7}{2}$ 2 4

Newton bölünmüş farklar interpolasyon polinomu

$$P(x) = \frac{x-x_1}{x_0-x_1}f(x_0) + \frac{x-x_0}{x_1-x_0}f(x_1)$$

$$= x\left(\frac{f(x_0)}{x_0-x_1} + \frac{f(x_1)}{x_1-x_0}\right) + \frac{f(x_0)x_1}{x_1-x_0} - \frac{f(x_1)x_0}{x_1-x_0}$$

$$f(x) = x\left(\frac{f(x_1)-f(x_0)}{x_1-x_0}\right) + \frac{f(x_0)x_1 - f(x_1)x_0}{x_1-x_0}$$

x	$f(x)$
x_0	$f(x_0)$
x_1	$f(x_1)$
$\vdots$	$\vdots$
x_{n-1}	$f(x_{n-1})$
x_n	$f(x_n)$

$P_n(x_k) = f(x_k)$, $k = 0, 1, 2, \dots, n$

$P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + \cdots + a_n(x-x_0)\cdots(x-x_{n-1})$

1. bölünmüş fark, 2. bölünmüş fark, k. bölünmüş fark

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$\cdots$	$f[x_i, x_{i+1}, \dots, x_{i+k}]$
x_0	$f(x_0)$	$\frac{f[x_1]-f[x_0]}{x_1-x_0}$	$f[x_0,x_1]$		
x_1	$f(x_1)$	$\frac{f[x_2]-f[x_1]}{x_2-x_1}$	$\frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$		
x_2	$f(x_2)$	$\frac{f[x_3]-f[x_2]}{x_3-x_2}$	$\frac{f[x_2,x_3]-f[x_1,x_2]}{x_3-x_1}$	$\frac{f[x_1,x_2,x_3]-f[x_0,x_1,x_2]}{x_3-x_0}$	
x_3	$f(x_3)$	$\frac{f[x_4]-f[x_3]}{x_4-x_3}$			
x_4	$f(x_4)$	$\frac{f[x_5]-f[x_4]}{x_5-x_4}$			
x_5	$f(x_5)$				

$$= \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, x_{i+1}, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$
$$+ f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

Örnek

x	$f(x)$
$-0,1$	$5,3$
0	2
$0,2$	$3,2$

a) NBFI 2. dereceden yapınız

b) $f(0,3) = 1$ verisi ile yeni 3. dereceden NBFI p

x	$f(x)$	$f[\,,\,]$	$f[\,,\,,\,]$	
$-0,1$	$5,3$	$\frac{3,3}{0,1} = -33$		
0	2	$\frac{1,2}{0,2} = 6$	$\frac{39}{0,3} = 130$	$\frac{-250}{3} + 130 = -\frac{1675}{3}$
$0,2$	$3,2$	$\frac{-2,2}{0,1} = -22$	$\frac{-25}{0,3} = \frac{-250}{3}$	$0,4$
$0,3$	1			

$$P_2(x) = 5,3 + (-33)(x + 0,1) + 130(x + 0,1)(x - 0)$$

$$= 130x^2 - 20x + 2$$

$$P_3(x) = P_2(x) + \left(-\frac{1675}{3}\right)(x + 0,1)(x - 0)(x - 0,2)$$

Uygu.

1) $f(x)=\sqrt{1+x}$ fonksiyonu ve $x_0=0$, $x_1=0,6$, $x_2=0,9$ düğüm noktaları verilmiş olsun. Aralık $[0, 0,9]$ olur.

a) 2. dereceden Lagrange interpolasyon polinomu belirleyiniz

b) Bu polinomu kullanarak $f(0,45)$'in tahmini değerini bulunuz. Hatasını değerlendiriniz.

a) $P_2(x) = L_{2,0}(x) f(x_0) + L_{2,1}(x) f(x_1) + L_{2,2}(x) f(x_2)$

$L_{2,0}(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$; $L_{2,1}(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$

$L_{2,2}(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$

$L_{n,k}(x) = \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{(x-x_i)}{(x_k-x_i)}$

$P_2 = \frac{1}{0,54}(x^2-0,15x+0,54) + \frac{1,2649106}{(-0,17)}(x^2-0,9x) + \frac{1,3764585}{0,27}(x^2-0,6x)$

b) $f(0,45) \approx P_2(0,45)$; $R_2(x) = \frac{f'''(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2)$